Standard Operating Procedure: Flood & Flash Flood Management (Part 1)

Issuing Authority: Central Water Commission (CWC) & NDMA, Government of India | **Document Reference:** CWC-NDMA-FL-2021-V1

Section 1.1: Riverine and Urban Flood Dynamics

Floods in India result from extreme monsoon precipitation, river channel overtopping, cloudbursts, and urban stormwater drain blockages. Urban flooding develops rapidly due to high impervious surface fractions, causing street waterlogging within minutes.

The Central Water Commission operates over 325 river forecasting stations nationwide. Flood warning levels are categorized into Warning Limit, Danger Limit, and Highest Flood Level (HFL).

Section 1.2: Pre-Monsoon Household Preparedness

Residents in low-lying and river basin flood plains must take proactive measures prior to the monsoon season:

- Keep vital documents (property titles, insurance, Aadhaar cards, medical history) stored in a certified waterproof floating pouch.
- Clean and unblock neighborhood rainwater drains and roof gutter pipes to prevent water accumulation.
- Identify elevated public shelters, municipal schools, or concrete multi-story community buildings designated by the District Disaster Management Authority (DDMA).
- Store drinking water purification chlorine tablets (Halazone) and oral rehydration salts (ORS) to mitigate waterborne gastrointestinal epidemics.

Standard Operating Procedure: Flood & Flash Flood Management (Part 2)

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Section 2.1: The 'Turn Around, Don't Drown' Motorist Protocol

Over 55 percent of all flood-related deaths occur inside vehicles when motorists underestimate the hydrodynamic force of moving water.

- Just 15 cm (6 inches) of rapidly flowing floodwater exerts sufficient lateral force to sweep an adult pedestrian off their feet.
- Just 30 cm (12 inches) of moving water will float and displace most passenger sedans, hatchbacks, and two-wheelers.
- Just 60 cm (24 inches) of water will sweep away heavy commercial trucks, buses, and sport utility vehicles (SUVs).
- NEVER drive through waterlogged roads, flooded dips, or submerged bridges. The asphalt underneath may have washed away into an invisible sinkhole.
- If your vehicle stalls in rising floodwater, abandon it immediately while doors can still be opened against water pressure, and proceed on foot to higher ground.

Section 2.2: Submerged Vehicle Extraction Technique

If exterior water pressure jams vehicle doors shut, occupants must take immediate window breach actions:

- Electric power windows must be rolled down immediately before water shorts the automotive electrical circuitry.
- If electric windows fail, remove a front car seat headrest and use its two pointed steel prongs to strike the lower corner of a SIDE window. Side windows are tempered glass and will shatter cleanly.
- Never attempt to smash the front windshield, as it is composed of shatterproof laminated safety glass.
- Push broken glass fragments outward with your footwear and climb out through the window frame onto the vehicle roof to wait for boat rescue.

Standard Operating Procedure: Flood & Flash Flood Management (Part 3)

Issuing Authority: Central Water Commission (CWC) & NDMA, Government of India | **Document Reference:** CWC-NDMA-FL-2021-V3

Section 3.1: Indoor Inundation & Vertical Evacuation

When floodwaters enter a residential structure, execute vertical relocation to upper floors:

- Disconnect the main electrical circuit breaker immediately if standing on a dry elevated platform. NEVER touch electrical switches or distribution boxes if standing in water or if the box itself is wet.
- Move all family members, portable emergency go-bags, and drinking water containers to the first floor or rooftop.
- If forced to take refuge on the roof, ensure access to the open sky. DO NOT remain enclosed inside an attic or loft without a pre-existing exterior roof exit hatch; rising water can turn closed attics into drowning traps.
- Signal rescue helicopters and boats by displaying a brightly colored cloth, orange tarpaulin, or white bedsheet on the rooftop.

Section 3.2: Health, Sanitation, and Water Purification

Post-flood disease outbreaks frequently cause more casualties than the initial inundation.

- All drinking water sourced from taps, wells, or rainwater must be brought to a rolling boil for at least 1 minute before consumption.
- Never consume fruits, vegetables, or food supplies that have had direct contact with floodwaters.
- Inspect living quarters for displaced venomous snakes, scorpions, and reptiles that seek shelter on dry furniture and roof rafters during floods.
- Wash skin exposed to floodwater with antiseptic soap to prevent leptospirosis and fungal infections.